Beginner Course on Microbial Genomics: Applications & Research (2026)

Module 2.2

Long-Read Sequencing
Oxford Nanopore Technology

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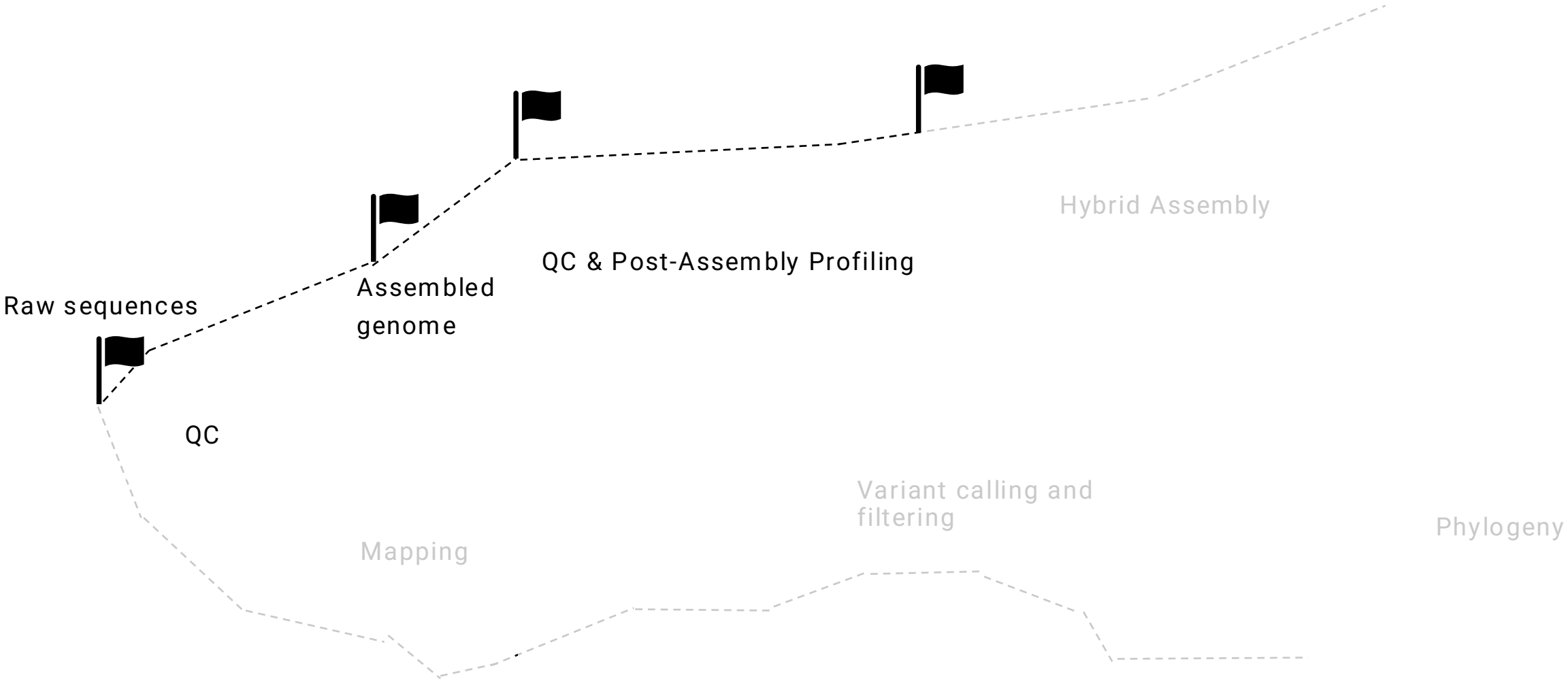
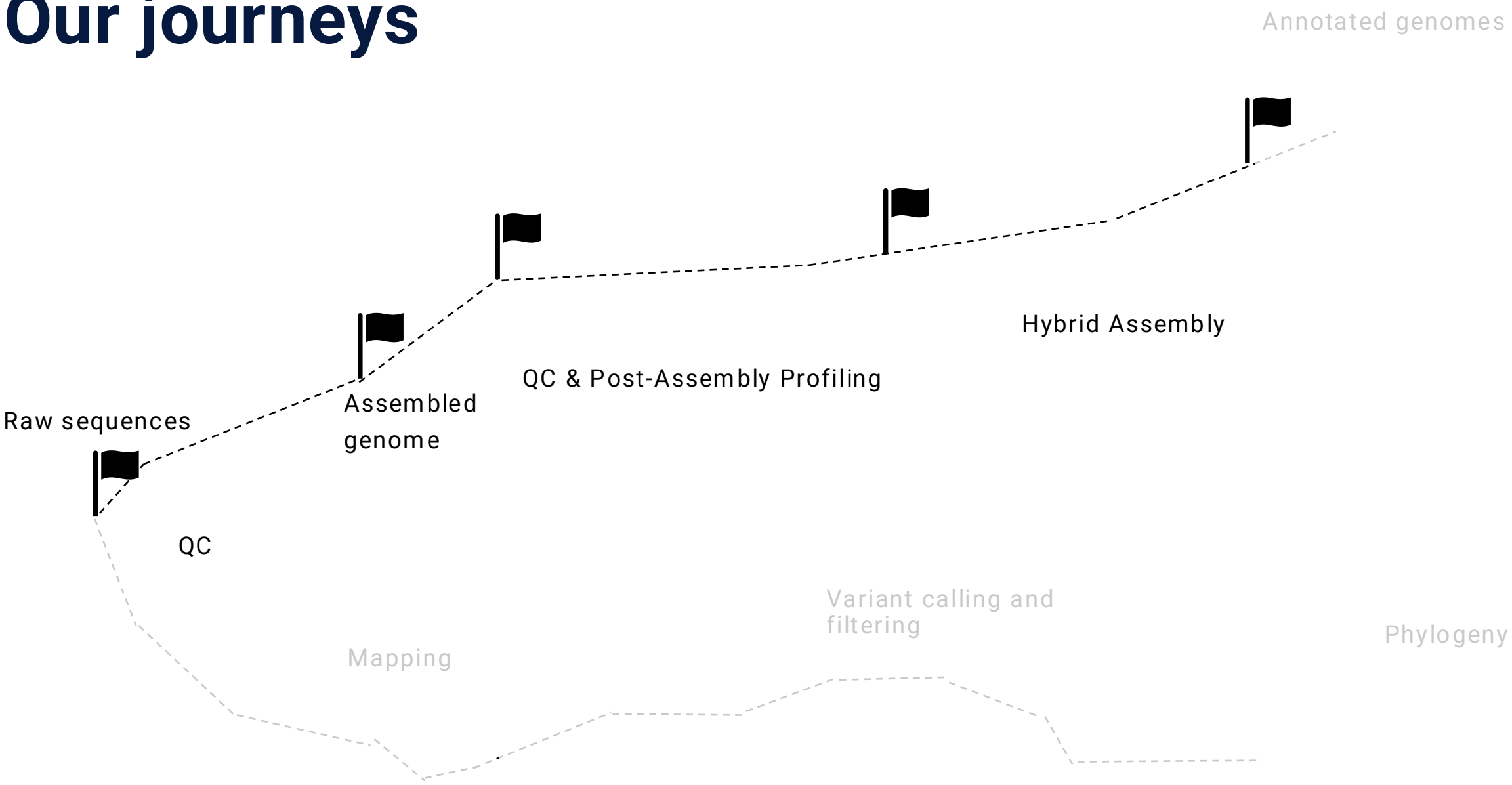
Our journeys



Our journeys



Agenda

I. **Theory**

1. Long-Read Introduction
2. Techniques and Platforms
3. Wet and Dry lab workflows
4. Long-Read vs Short-Read Sequencing

II. **Practical Command-line (CLI)**

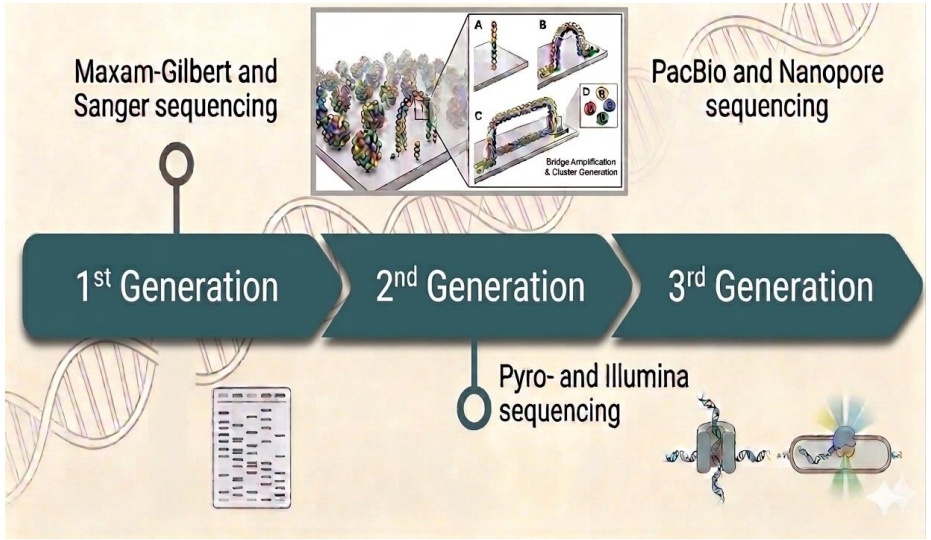
Unicycler

III. **Data Visualization and Manipulation with Bandage**

I. Theory

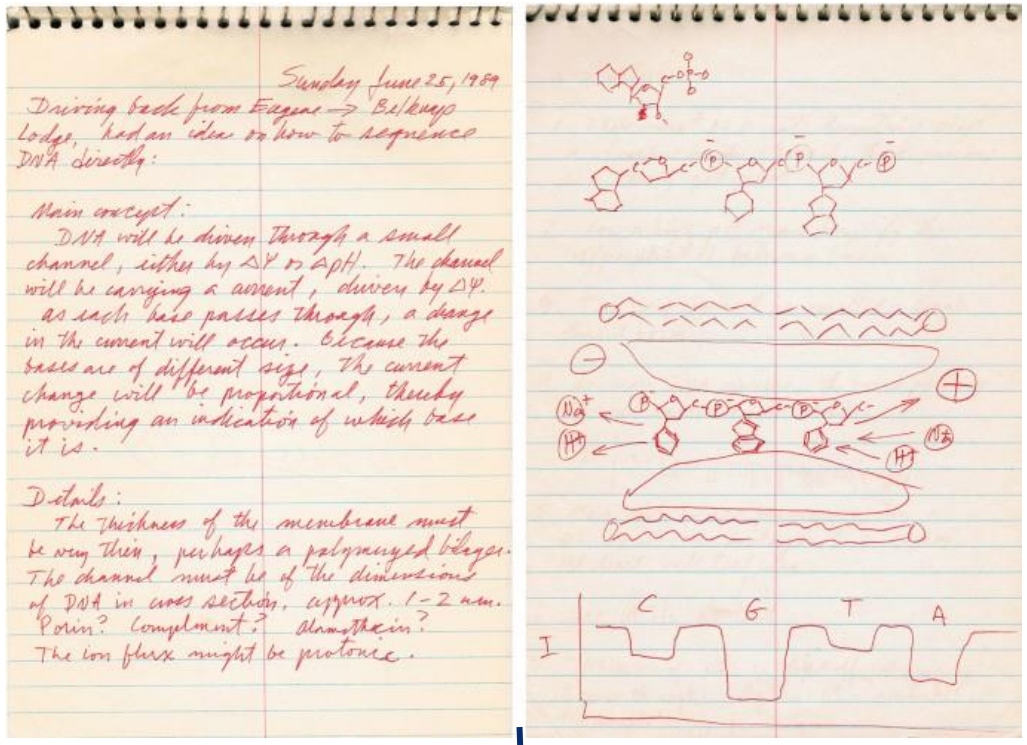
Theory

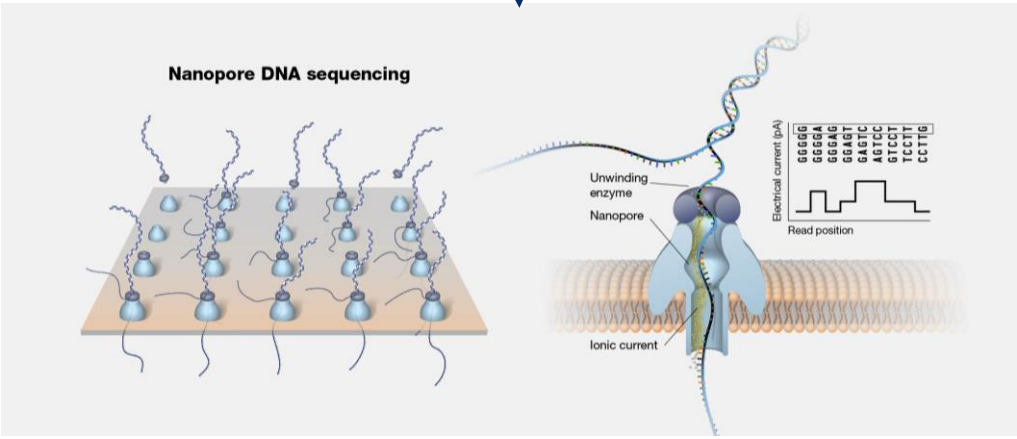
Long-Read Introduction



History of DNA sequencing

David Deamer's original notebook sketch, 1989





Theory

Techniques and Platforms

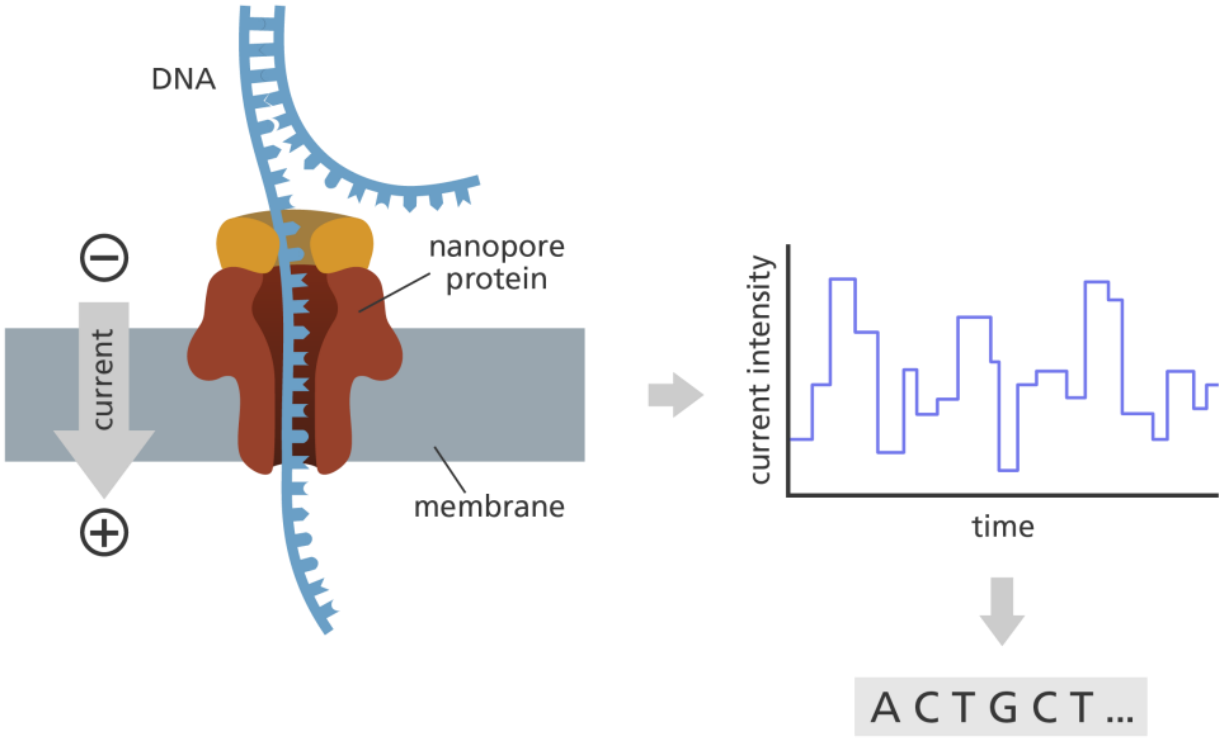


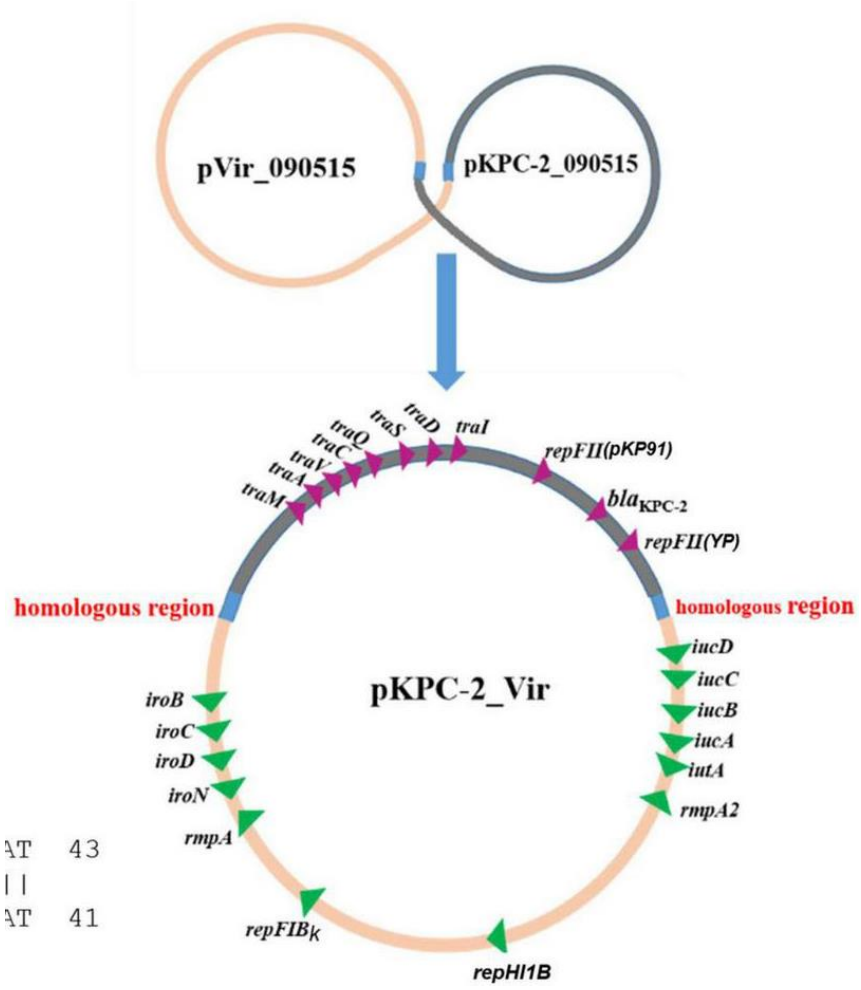
Illustration of nanopore
(https://www.goldengooseaward.org/)

Theory

Why do we need Long-Read?

- Faster and simpler workflows (Outbreak response – Ebola, COVID, Hantavirus,…)

- Complete bacterial genome (AMR-carrying plasmid, VF-carrying plasmid,…), plasmid population diversity



10.3389/fmicb.2022.852596

Theory

Long-Read vs Short-Read Sequencing

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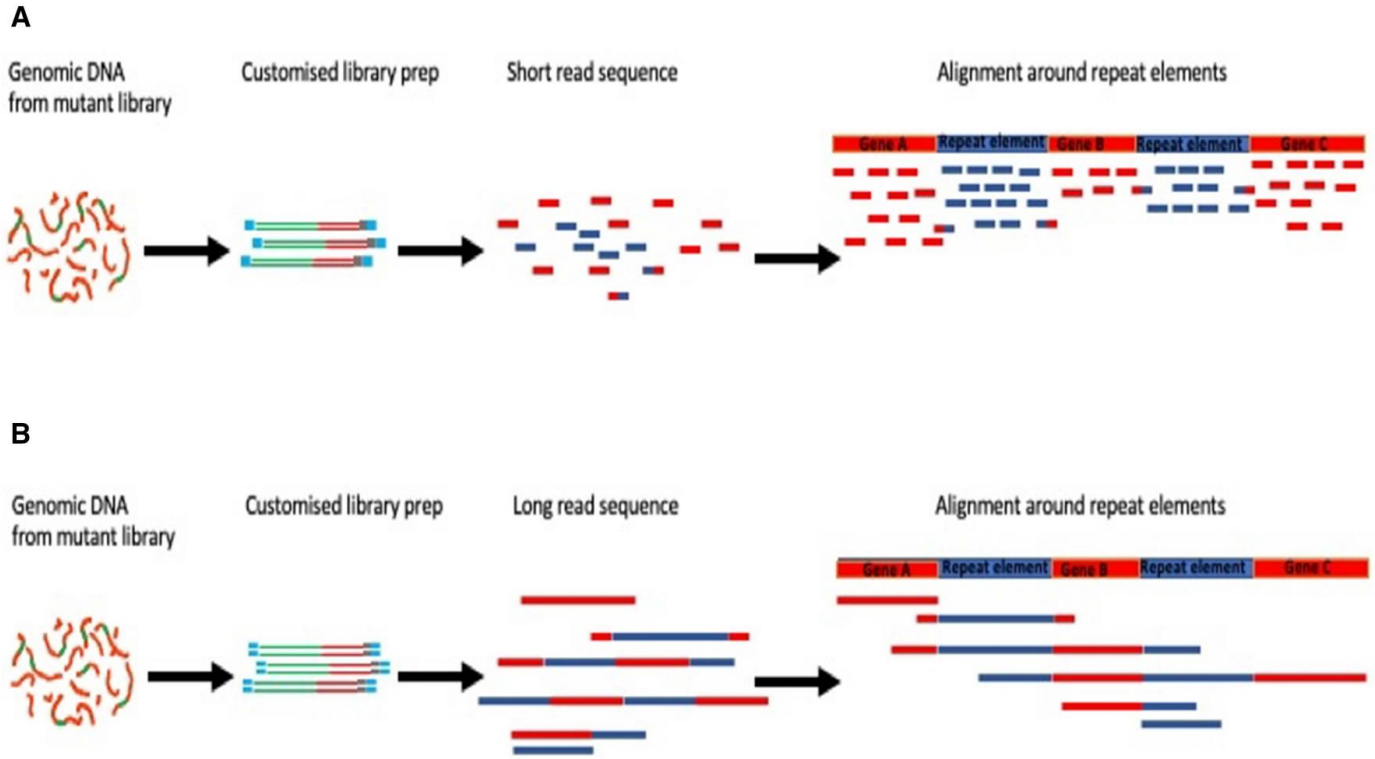
Theory

Long-Read vs Short-Read Sequencing



10.1038/s41598-022-07557-x

Feature	**Short-Read**	**Long-Read**
Read length	100–300 bp	5 kb → Mb
Raw accuracy	~99.9%	~95–98%
Cost per Gb	low	low–medium
Throughput	very high	scalable
Flexibility	low	high
Real-time	no	yes

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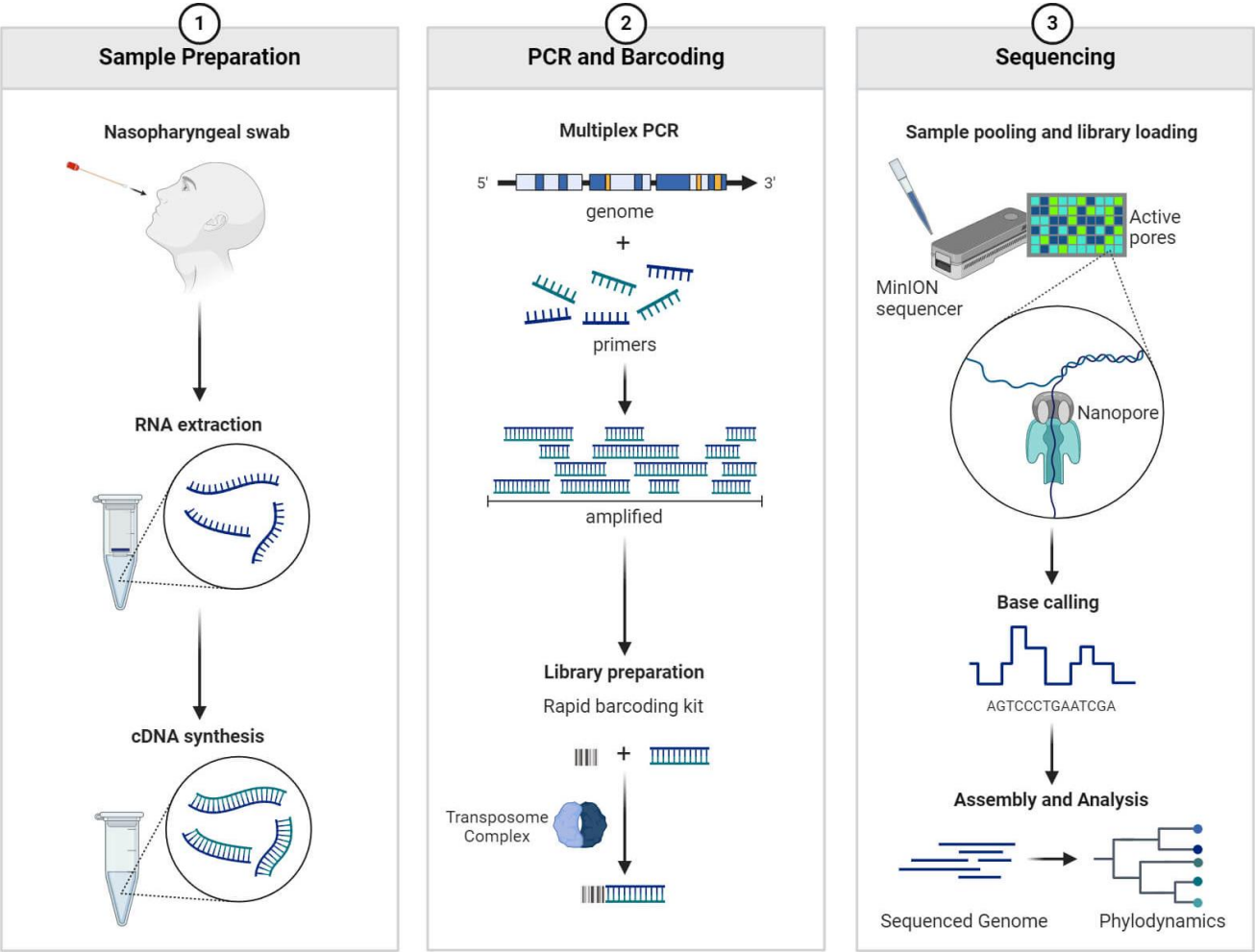
Hybrid assembly

Theory

Wet and Dry lab workflows

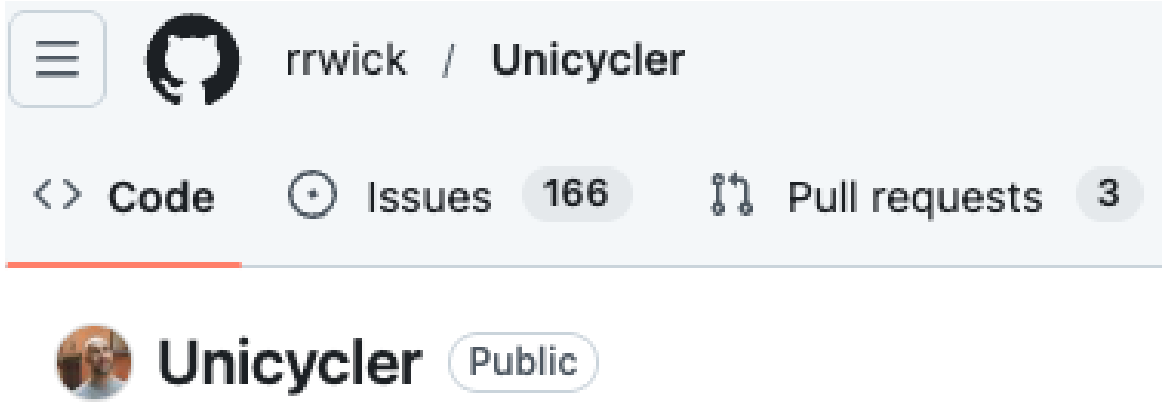


10.1515/ntrev-2022-0036

II. Practical Command-line (CLI)

Hybrid Assembly – unicycler

https://github.com/rrwick/unicycler



III. Data Visualization and Manipulation

Bandage / BandageNG

THE END.

Thank you for your attention

Module 2.2: Long-Read Sequencing - Hybrid Assembly